

MemeCLIP: Leveraging CLIP Representations for Multimodal Meme Classification

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Abstract

The complexity of text-embedded images presents a formidable challenge in machine learning given the need for multimodal understanding of the multiple aspects of expression conveyed in them. While previous research in multimodal analysis has primarily focused on singular aspects such as hate speech and its subclasses, our study expands the focus to encompass multiple aspects of linguistics: hate, target, stance, and humor detection. We introduce a novel dataset PrideMM comprising text-embedded images associated with the LGBTQ+ Pride movement, thereby addressing a serious gap in existing resources. We conduct extensive experimentation on PrideMM by using unimodal and multimodal baseline methods to establish benchmarks for each task. Additionally, we propose a novel framework MemeCLIP for efficient downstream learning while preserving the knowledge of the pre-trained CLIP model. The results of our experiments show that MemeCLIP achieves superior performance compared to previously proposed frameworks on two real-world datasets. We further compare the performance of MemeCLIP and zero-shot GPT-4 on the hate classification task. Finally, we discuss the shortcomings of our model by qualitatively analyzing misclassified samples.

1 Introduction

In recent years, the pervasive integration of social media platforms into everyday life has resulted in an exponential increase in the generation and dissemination of multimedia content. At the heart of this digital ecosystem lies the meme: a text-embedded image imbued with humor, wit, and often, a subversive edge, which offers a medium through which individuals can express opinions, share experiences, and engage in online activism (Moreno-Almeida, 2021; Baker et al., 2020). With their ability to distill complex ideas into digestible units of communication, memes have emerged as a

powerful medium for expressing both support and opposition toward socio-political events (Imperato et al., 2023).

However, with opinions being expressed freely, hate speech becomes prevalent, often targeted towards individuals, organizations, and even marginalized communities (Thapa et al., 2023), targeting them with vitriol and prejudice (Lingiardi et al., 2020; Imperato et al., 2023). Particularly, the LGBTQ+ movement stands as a prominent exemplar of online discourse, where memes serve as vehicles of both solidarity and resistance, reflecting the multifaceted dynamics of attitudes and perceptions within the community and beyond (Gal et al., 2016). In this context, the distinction between humor and harm becomes blurred, as memes straddle the line between satire and offense, challenging researchers and platforms alike to navigate the complexities of online content moderation (Langvardt, 2017). Previous attempts that endeavored to suppress such content have resulted in the discriminative suppression of all LGBTQ+ content (Griffin, 2022, 2024), which can harm the awareness and acceptance of this community. Thus, understanding the nuances of hate speech, opinions, and intended humor within memes is paramount for fostering an inclusive digital environment and combating online discrimination.

To address these challenges, we introduce PrideMM: a novel dataset comprising 5,063 text-embedded images related to the LGBTQ+ movement annotated with a multi-aspect schema encompassing four tasks:

- **Task A:** Detection of Hate Speech
- **Task B:** Classifying the Targets of Hate Speech
- **Task C:** Classification of Topical Stance
- **Task D:** Detection of Intended Humor

The analysis of text-embedded images is particularly challenging given the need for contextual understanding and the prevalence of ambiguity and subjectivity in them (Sherratt, 2022).

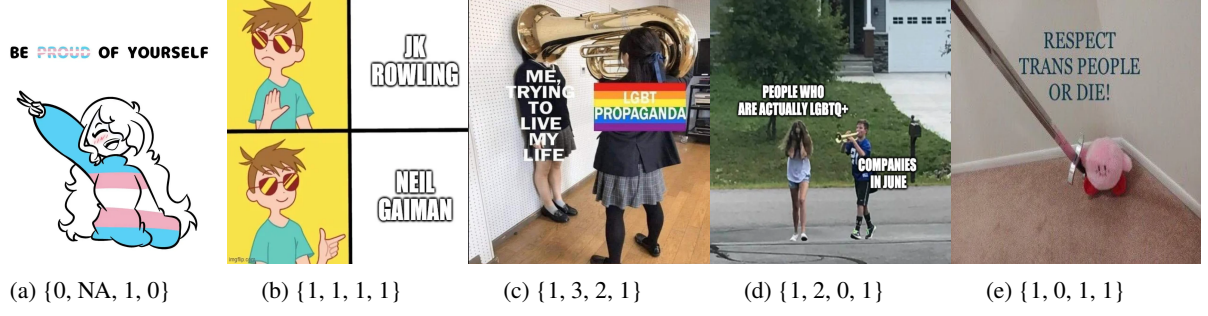


Figure 1: Examples of text-embedded images labeled for all aspect labels in the PrideMM dataset. The labels are in the form of {Hate, Target, Stance, Humor}. For Hate, {0, 1} correspond to *No Hate* and *Hate* respectively. For Target, {0, 1, 2, 3} correspond to hate targeted towards *Undirected*, *Individual*, *Community*, and *Organization* respectively. For Stance, {0, 1, 2} correspond to *Neutral*, *Support*, and *Oppose* respectively. For Humor, {0, 1} correspond to *No Humor* and *Humor* respectively.

The multi-aspect nature of our dataset provides a more holistic view of the diverse themes usually expressed through memes. Samples of images and their annotations are illustrated in Figure 1. Through PrideMM, we aim to cultivate a more profound understanding of interaction dynamics through memes on social media and facilitate the development of multimodal content moderation methods in order to make the internet a safer space. We implement a range of baseline and state-of-the-art hate speech detection models to establish benchmarks for each task of PrideMM.

We further propose MemeCLIP, a novel framework that leverages the knowledge of CLIP (Contrastive Language-Image Pre-Training) (Radford et al., 2021) for multi-aspect meme classification. We employ linear layers to effectively disentangle image and text representations in CLIP’s multimodal embedding space. We utilize lightweight Feature Adapters to preserve the prior knowledge of CLIP and adapt its embedding spaces to the meme classification task while avoiding overfitting on smaller datasets. We further integrate a cosine classifier alongside Semantic-Aware initialization (Shi et al., 2023) to make MemeCLIP more robust to the imbalanced class labels that may exist in datasets such as PrideMM and HarMeme (Pramanick et al., 2021a) that are representative of real-world data distributions. Distinct from previous multimodal meme classification frameworks, MemeCLIP is trained end-to-end in a single step and does not rely on extraneous models to create augmented data. Our main contributions can be summarized as follows:

- We release PrideMM, a dataset containing 5,063 text-embedded images related to the LGBTQ+ movement.
- We benchmark PrideMM by using various uni-

modal and multimodal methods including previously proposed multimodal frameworks for hate speech detection.

- We introduce MemeCLIP, a novel framework that utilizes lightweight modules on top of a frozen CLIP model to classify memes.

2 Related Work

2.1 Multimodal Datasets

Multimodal image-text analysis has seen significant strides in recent years, owing to its widespread popularity and high availability across social media platforms. Qu et al. (2022) introduced the DisinfoMeme dataset that contains memes from three different contexts: the COVID-19 pandemic, the Black Lives Matter (BLM) movement, and Veganism, labeled for disinformation identification. In an effort to discern humor often expressed in memes, Tanaka et al. (2022) created a humor detection dataset by proposing a refined process for extracting memes devoid of interpersonal influence on the perception of humor. With the increasing need for hate speech and offensive content detection in social media, multimodal data analysis for hate speech has seen a particular surge. The HMC dataset (Kiela et al., 2020) was released as part of the Hateful Meme Challenge and contains synthetic memes designed to convey contrastive meanings from image and text modalities. Hateful memes in the dataset target religion, race, disability, and sex. Suryawanshi et al. (2020) employed extensive annotation guidelines to create the MultiOFF dataset, consisting of memes collected from Reddit, Facebook (FB), Twitter, and Instagram for offensive content detection. The Harm-C (Pramanick et al., 2021a) and Harm-P (Pramanick et al.,

Work	Data Source	Multimodal	Sub-Classes	Multi-aspect	Size	Context
Qu et al. (2022)	Reddit	✓	✗	✗	1,170	COVID-19, BLM, Veganism
Tanaka et al. (2022)	Meme Websites	✓	✗	✗	7,500	General Discourse
Kiela et al. (2020)	Self-Generated	✓	✗	✗	10,000	General Discourse
Suryawanshi et al. (2020)	FB, Twitter, Instagram	✓	✗	✗	743	U.S. Election
Pramanick et al. (2021a)	Google Images	✓	✓	✗	3,544	COVID-19
Pramanick et al. (2021b)	Google Images	✓	✓	✗	3,522	U.S. Politics
Bhandari et al. (2023)	Twitter, FB, Reddit	✓	✓	✗	4,723	Russia-Ukraine War
Dacon et al. (2022)	Reddit	✗	✓	✓	9930	LGBTQ+ Movement
Gautam et al. (2020)	Twitter	✗	✗	✓	9937	#MeToo Movement
Ousidhoum et al. (2019)	Twitter	✗	✓	✓	13,000	General discourse
PrideMM (Ours)	FB, Twitter, Reddit	✓	✓	✓	5,063	LGBTQ+ Movement

Table 1: Summary of datasets used in the literature.

2021b) datasets comprise memes related to the COVID-19 pandemic and US politics respectively. Both datasets were annotated for three degrees of harmfulness and four subclasses for the targets of hate speech. Similarly, Bhandari et al. (2023) collected text-embedded images from social media platforms - Twitter, Facebook, and Reddit related to the Russia-Ukraine Conflict for hate speech detection, annotating samples for hate speech detection and target detection.

2.2 Multi-aspect Datasets

Online discourse on socio-political events is often imbued with a series of human emotions, which has led researchers to study the multiple aspects of linguistics expressed in them. Dacon et al. (2022) used comments collected from RedditBias (Barik-eri et al., 2021) related to LGBTQ+ individuals and annotated each comment for the presence of Toxicity, Severe Toxicity, Obscene, Threat, Insults, and Identity Attacks. Similarly, Gautam et al. (2020) curated a dataset of tweets related to the #MeToo movement in social media by annotating the tweets in five different aspects. Taking the studies of multiple aspects one step further, Ousidhoum et al. (2019) compiled an extensive multi-aspect Twitter dataset in English, French, and Arabic with each language dataset annotated for different aspects including hate and offensiveness. Table 1 provides a detailed comparison of the datasets used in the literature.

Multi-aspect data helps better understand the spectrum of human emotions that may be associated with social media interactions. Most multimodal datasets, while only focusing on a single aspect and its sub-classes, fail to encompass the complex dynamics of emotions expressed by the masses. Our work aims to address this gap by presenting a multimodal and multi-aspect dataset

comprising three different aspects- hate detection, stance detection, and humor detection, and one subclass within hate- target detection, to enable more nuanced studies of multimodal meme data through computational methods.

2.3 Multimodal Frameworks

Recent developments in large vision-language models have incited a wave of research in methods to tackle hate speech in text-embedded images. MOMENTA (Pramanick et al., 2021b) was one of the first frameworks proposed to incorporate CLIP’s vision and language encoders for multimodal hate speech classification. They further extract regions of interest from image data and named entities from text data and combine them with CLIP representations by using cross-modal attention fusion. Similarly, HateCLIPper (Kumar and Nandakumar, 2022) was proposed to better model cross-modal interactions between CLIP representations. Textual inversion (Gal et al., 2022) has been used to integrate visual cues in the text representation space in frameworks such as ISSUES (Burbi et al., 2023). Recent works make use of image caption models to extract text captions from images and learn a single language processing model (Cao et al., 2023b,a). However, rather than relying on augmented data from extraneous models, our proposed framework MemeCLIP leverages the knowledge learned by CLIP’s encoders during its pre-training step to process the rich multimodal information inside each image. Additionally, we use Feature Adapters alongside residual connections to prevent overfitting as annotated datasets for multimodal meme classification generally lack a high number of samples. We further utilize a cosine classifier to make MemeCLIP more robust to imbalanced data classes, which is prevalent in multi-label tasks in this domain.

3 Dataset

In this section, we describe various aspects of our dataset including data collection, annotation guidelines, and dataset statistics. Our dataset comprises 5,063 text-embedded images that encompass memes, posters, and infographics relevant to the LGBTQ+ movement. We include images from 2020-2024 as this period saw an upsurge of social media content in this domain (Oz et al., 2023). This also allows our dataset to represent contemporary social media interactions through memes. Note that by the term LGBTQ+, we refer to all gender identities and sexual orientations.

3.1 Data Collection

To maintain diversity in the dataset, we collected data from three popular social media platforms- Facebook, Twitter, and Reddit, through manual search and extraction. For Twitter, we used hashtags such as *#lgbt*, *#pride*, *#trans*, *#transrights*, *#nonbinary*, and *#genderidentity* to filter images related to LGBTQ+ discussions. For Facebook, we targeted groups that frequently discussed LGBTQ+ content. Similarly, for Reddit, we identified subreddits where discussion related to LGBTQ+ was more prominent. Further, to ensure the relevance and quality of the dataset, the data collection process was subject to filtering criteria. Detailed filtering criteria for our dataset can be found in Appendix A.2. Since different annotators may collect duplicates of the same image, we sequentially employed two image deduplication tools, *dupeGuru*¹ and *diffPy*², to filter out duplicates and retain the highest quality images out of the batch of duplicates. We used the OCR application provided by Google Cloud Vision API³ to extract textual data from the images. We removed non-alphanumeric elements such as special characters, hyperlinks, symbols, and non-English characters to reduce noisy text data and ensure data quality. Note that the text may occasionally contain unintentional noisy artifacts.

3.2 Data Annotation

We engaged five experienced annotators, well-versed in NLP and computational linguistics, to annotate data samples for PrideMM. The annotators had a prior understanding of the LGBTQ+ move-

ment and meme dynamics on social media. We presented them with comprehensive annotation guidelines to ensure uniform and unbiased annotations and asked them to annotate each image separately for all four tasks. A 3-phase annotation schema was used to ensure accurate and consistent annotations. First, a dry run was conducted where the annotators were given 50 images each for annotation to evaluate the understanding of the annotation guidelines among the annotators. Second, a revision phase was conducted where the annotators were given another 200 images each and received a revised set of instructions based on the results of the first phase. Finally, in the consolidation phase, the annotators annotated 50 images while discussing and revising the annotation guidelines until a consensus was reached. These steps were taken to avoid misannotations and noisy labels in PrideMM as much as possible. We followed meticulously devised annotation guidelines to ensure consistency in the annotations. Every image in our dataset was annotated for three different aspects and one subclass. Apart from the relationship between 'Hate' and 'Hate Targets', all aspects were annotated independently. Detailed annotation guidelines can be found in Appendix A.3.

3.3 Statistics and Inter-Annotator Agreement

Table 2 shows the distribution of images in PrideMM across all class labels. For the hate detection task, the dataset has a balanced distribution of binary labels. The target detection task exhibits an imbalanced distribution. Given the context of this study, most hateful images convey undirected hate or are targeted toward communities, with low occurrences of hate against individuals and organizations. For the stance detection task, the number of images is well-balanced along three labels. On the other hand, as memes are often meant to be humorous, the majority of the images in the dataset are humorous. We provide a detailed analysis of PrideMM's text content by using topic modeling in Appendix A.4.

We used the Fleiss' Kappa (κ) (Faloutico and Quatto, 2015) as a statistical measure to assess the inter-annotator agreement across all four tasks. For Task A (Hate Speech detection), the Fleiss' (κ) was 0.66/0.74 in the dry run and final phase respectively, for Task B (Target detection), κ was 0.68/0.81, for Task C (Stance detection), κ was 0.62/0.75, and for Task D (Humor detection), κ was 0.60/0.74. The increase in the κ score from the dry run phase to the

¹<https://github.com/arsenatar/dupeguru>

²<https://github.com/elisemercury/Duplicate-Image-Finder>

³<https://cloud.google.com/vision/docs>

final phase across all tasks reflects the effectiveness of the 3-phase annotation schema.

Task	Label	#Samples	%
Hate	No Hate	2,581	50.97%
	Hate	2,482	49.03%
Target	Undirected	771	31.07%
	Individual	249	10.03%
	Community	1,164	46.90%
	Organization	298	12.00%
Stance	Neutral	1,458	28.50%
	Support	1,909	37.70%
	Oppose	1,696	33.50%
Humor	No Humor	1,642	32.43%
	Humor	3,421	67.57%

Table 2: Dataset Statistics for PrideMM. The data consists of 5,063 samples for Hate, Stance, and Humor classification tasks, and 2,482 samples for the Target classification task.

4 Methodology

In this section, we describe our proposed framework, MemeCLIP, for multi-aspect meme classification. We utilize the vision-language model CLIP (Radford et al., 2021) to create rich representations that effectively encapsulate the semantics of a meme. We add lightweight modules on top of CLIP to disentangle image and text representations, prevent overfitting, and make MemeCLIP more robust to imbalanced data. Figure 2 illustrates the overall architecture of MemeCLIP. Below, we describe each component of MemeCLIP in detail.

Zero-shot CLIP. The vision-language model CLIP exhibits stellar zero-shot performance and transfer learning capabilities (Radford et al., 2021; Wang et al., 2023). CLIP is pre-trained on 400 million image-text pairs from the internet, enabling it to encode visual and textual data in a shared embedding space. The model consists of an Image Encoder E_I and a Text Encoder E_T . We freeze the weights of both encoders to preserve the valuable knowledge captured by them during pre-training. The unimodal image and text representations $F_I, F_T \in \mathbb{R}^{768}$ effectively encapsulate the semantics of a meme and are defined as:

$$F_I = E_I(I); F_T = E_T(T) \quad (1)$$

where I is the image and T is its text pair.

Linear Projection Layers. While the contrastive pre-training objective of CLIP promotes similarity between corresponding text and image pairs, memes often involve contrastive visual and

linguistic content to evoke a sense of irony. Similar to (Kumar and Nandakumar, 2022), we employ individual linear projection layers for each modality to effectively disentangle image and text representations in the shared embedding space. These projection layers result in the unimodal projections $F_I^{proj}, F_T^{proj} \in \mathbb{R}^{1024}$, mapping the representations to the dimensions of CLIP’s last hidden state, $D_{CLIP} \in \mathbb{R}^{1024}$, which enables the use of Semantic-Aware Initialization.

$$F_I^{proj} = L_I^{proj}(F_I); F_T^{proj} = L_T^{proj}(F_T) \quad (2)$$

where L_I^{proj} and L_T^{proj} represent the image and text projection layers respectively.

Feature Adapters. Since CLIP is pre-trained on an extensive amount of data, it may exhibit symptoms of overfitting when applied to smaller datasets for downstream tasks. Inspired by (Gao et al., 2024), we adopt lightweight Feature Adapters for both image and text modalities to learn the features of the new data while retaining CLIP’s prior knowledge. We further utilize residual connections to integrate prior image and text projections to the outputs of the Adapters, allowing our model to balance the knowledge of the fine-tuned adapter and the disentangled image and text projections. We use a residual ratio α to maintain harmony between these two modules. With the image and text Feature Adapters A_I and A_T respectively, the final unimodal representations $F_I, F_T \in \mathbb{R}^{1024}$ are obtained as follows:

$$F_I = \alpha A_I(F_I^{proj}) + (1 - \alpha) F_I^{proj} \quad (3)$$

$$F_T = \alpha A_T(F_T^{proj}) + (1 - \alpha) F_T^{proj} \quad (4)$$

Modality Fusion. Owing to the extensive unimodal feature modeling, MemeCLIP avoids the need for trainable fusion layers like Cross-Modal Attention Fusion in MOMENTA (Pramanick et al., 2021b) and Combiner in ISSUES (Burbi et al., 2023). We fuse the image and text representations by using an element-wise multiplication operation ($\odot$) to obtain a single multimodal representation $F_{MM} \in \mathbb{R}^{1024}$. We further alleviate the need for the two-stage training process employed in HateCLIPper (Kumar and Nandakumar, 2022) and ISSUES.

$$F_{MM} = F_I \odot F_T \quad (5)$$

F_{MM} is then passed through a linear pre-output layer before classification.

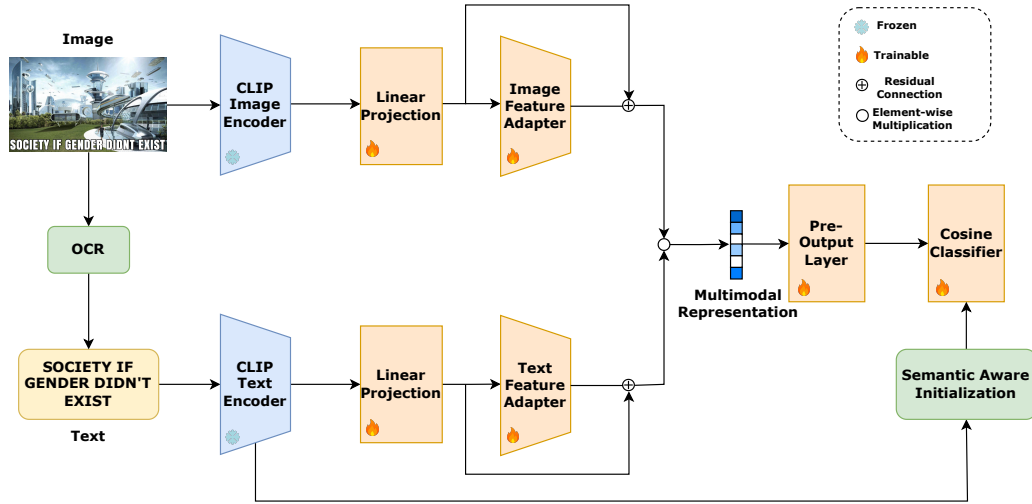


Figure 2: An overview of our proposed framework, MemeCLIP. We use frozen image and text CLIP encoders to create representations for each image-text pair. These representations are passed through linear layers to disentangle the modalities in CLIP’s shared embedding space. We implement Feature Adapters with residual connections for each modality to prevent overfitting. We use a cosine classifier to make MemeCLIP more robust to imbalanced data. We initialize classifier weights by using Semantic-Aware Initialization to further improve performance.

Classification. For classification, we employ a cosine classifier (Liu et al., 2020) that is more robust to biases in prediction under data imbalances. Following (Shi et al., 2023), we adopt Semantic-Aware Initialization (SAI) to initialize the weights of this classifier by exploiting the semantic knowledge within the text encoder of CLIP. We encode class labels by using the prompt "A photo of {LABEL}" into $F_{class} \in \mathbb{R}^{n \times 1024}$ where n is the number of classes, and use F_{class} to initialize the classifier weight W_{class} . During training, the predicted logit Z for a class x is calculated as follows:

$$Z_x = \sigma \times \frac{W_x \times F_{MM}}{\|W_x\|_2 \|F_{MM}\|_2} \quad (6)$$

where σ is a static scaling factor for the cosine classifier.

5 Experimental Results

Table 3 and Table 4 show our experimental results for the PrideMM and HarMeme datasets respectively. We pre-define train/validation/test splits in the ratio 85/5/10 respectively for PrideMM, and use the pre-defined split for the HarMeme dataset. We conduct experiments on unimodal and multimodal baseline methods and previous frameworks proposed for multimodal hate classification. We conduct each experiment on three random seeds and report the Mean and Standard Deviation ($\pm$) values for Accuracy, AUC (Macro), and F1-Score (Macro). We use ViT-L/14 as the image encoder for

all CLIP-based methods except for MOMENTA, which uses ViT-B/32 as the backbone for its CLIP model by default. For CLIP, we use concatenation to fuse unimodal feature representations. Further implementation details are outlined in A.1.

5.1 PrideMM Dataset

Unimodal Methods. For the unimodal methods, we used ViT-L/14 (Dosovitskiy et al., 2020) and CLIP’s image encoder as image-based methods, and BERT (Devlin et al., 2018) and CLIP’s text encoder as text-based methods. The image-based methods generally performed better than their text-based counterparts across all tasks, substantiating that Transformer-based visual models create meaningful representations that also capture the semantic meaning conveyed by the text embedded in the pixel space (Burbi et al., 2023). The text-based methods showed poor performance on the multi-label target and stance detection tasks. The image encoder of CLIP shows superior results compared to the standard pre-trained Visual Transformer while having the same architecture, demonstrating the effectiveness of contrastive pre-training. The unimodal methods generally perform worse than any multimodal method across all tasks, underscoring the need for multimodal processing in meme analysis.

Multimodal Methods. We tested the performance of the multimodal methods CLIP (Radford et al., 2021), CLIP-Adapter (Gao et al., 2024),

Method	Hate			Target			Stance			Humor		
	Acc.	AUC.	F1	Acc.	AUC.	F1	Acc.	AUC.	F1	Acc.	AUC.	F1
BERT	71.12±0.67	75.33±0.50	70.06±0.22	54.25±1.32	75.52±0.77	54.03±1.51	52.30±0.91	67.10±0.74	51.11±1.34	71.04±0.55	72.25±1.77	64.60±0.98
CLIP Text-Only	68.64±0.86	74.52±0.95	68.62±0.88	50.34±0.62	72.67±1.34	47.65±1.18	50.43±1.08	67.60±0.55	49.26±1.25	69.23±1.54	70.52±1.42	62.02±1.69
ViT-L/14	69.23±2.22	77.05±0.48	68.24±3.01	58.36±1.56	79.13±1.48	50.24±2.93	58.80±1.26	73.20±4.62	56.14±4.76	73.04±2.20	77.71±1.13	69.18±2.35
CLIP Img-Only	70.01±0.78	80.53±0.42	72.66±3.08	60.32±1.77	80.89±0.73	57.19±3.25	61.01±0.82	77.48±0.66	57.87±0.78	76.14±0.19	82.1±1.42	72.37±1.13
CLIP	72.39±1.20	80.47±0.61	72.33±1.26	61.14±0.59	81.92±0.44	58.46±1.02	59.31±0.82	76.92±0.87	57.81±1.14	76.66±1.32	80.73±0.20	73.23±1.56
CLIP-Adapter	72.75±1.09	80.91±0.56	72.69±1.02	61.59±0.52	82.14 ±0.35	58.08±0.91	59.55±0.47	77.23±0.73	57.93±0.91	77.01±1.01	80.97±0.71	73.51±0.97
MOMENTA	72.23±0.58	78.55±0.50	71.78±0.35	57.28±1.26	78.89±1.23	52.79±1.84	55.62±1.90	73.64±2.35	54.84±2.28	74.16±2.17	77.38±1.63	71.34±2.70
HateCLIPper	75.53±0.58	83.12±0.44	74.08±0.37	62.49±2.06	80.32±1.42	56.77±0.72	63.24 ±0.69	77.99±1.25	57.15±0.76	76.13±0.19	83.50±0.51	75.41±0.28
ISSUES	74.68±1.62	84.17±0.45	73.64±2.48	61.25±2.00	78.73±0.21	58.30±0.17	59.39±1.08	77.02±1.93	57.27±1.40	78.95±0.88	84.78±0.60	75.73±2.17
MemeCLIP	76.06 ±0.23	84.52 ±0.31	75.09 ±0.20	66.12 ±0.47	81.66±0.25	58.65 ±0.97	62.00±0.12	80.11 ±0.15	57.98 ±1.91	80.27 ±0.52	85.59 ±0.23	77.21 ±0.79

Table 3: Classification performance of models on the PrideMM dataset. The results are in the form of *Mean ± Standard Deviation*. Performance is reported across three evaluation metrics: Accuracy, AUROC (Macro), and F1-Score (Macro). The best performance is highlighted in **bold**.

MOMENTA (Pramanick et al., 2021b), HateCLIPper (Kumar and Nandakumar, 2022), ISSUES (Burbi et al., 2023), and our framework, MemeCLIP. MemeCLIP outperforms the baseline CLIP model and previously proposed multimodal methods across all metrics in the hate and humor classification tasks. Our model performs particularly well in the four-class target classification task which has merely half the number of samples as the other tasks and harbors a heavy class imbalance, demonstrating the model’s robustness to overfitting and data imbalance settings. Smaller frameworks with a lower number of parameters such as the baseline CLIP, CLIP-Adapter, HateCLIPper, and MemeCLIP perform optimally in this task with CLIP-Adapter showing the highest AUC, while the remaining methods show symptoms of overfitting. In the stance classification task, HateCLIPper surpasses MemeCLIP in accuracy, while the latter shows a higher AUC and F1-Score. While the CLIP-Adapter model is similar to MemeCLIP, it uses a single feature adapter as individual feature adapters for both modalities may carry redundant information from the unimodal encoders to the classifier; however, MemeCLIP outperforms this model in most tasks by using feature adapters for both image and text modalities, which may signify that separate layers for each modality are more effective at encoding memes that may contain visual and language content that convey different meanings individually and when put together.

5.2 HarMeme Dataset

To test the generalizability of MemeCLIP across other meme datasets, we perform all the experiments on the HarMeme dataset (Pramanick et al.,

Method	Acc.	AUC.	F1
BERT	71.05±0.70	76.34±0.48	68.83±0.49
CLIP Text-Only	73.79±0.23	79.31±0.81	71.60±0.21
ViT-L/14	77.27±1.71	85.53±0.41	75.55±2.63
CLIP Img-Only	79.38±0.25	88.54±2.53	78.66±0.11
CLIP	81.36±0.81	87.27±0.67	80.30±0.95
CLIP-Adapter	82.21±0.73	87.53±0.61	80.89±0.87
MOMENTA	82.44±0.65	87.88±0.37	81.49±0.45
HateCLIPper	83.68±0.62	90.83±0.46	83.31±0.41
ISSUES	81.31±1.05	91.98±0.61	80.45±0.87
MemeCLIP	84.72 ±0.45	92.07 ±0.34	83.74 ±0.43

Table 4: Classification performance of models on the HarMeme dataset. The results are in the form of *Mean ± Standard Deviation*. Performance is reported across three evaluation metrics: Accuracy, AUROC (Macro), and F1-Score (Macro). The best performance is highlighted in **bold**.

2021a), which consists of real-world hateful memes shared on social media in the context of COVID-19. Similar to previous studies, we find that the performance of the multimodal models surpasses the unimodal models’ performance by a decent margin. MemeCLIP outperforms the other multimodal frameworks in this dataset, which demonstrates its effectiveness over the state-of-the-art baselines.

5.3 Ablation Study

We conduct a systematic ablation study to assess the contribution of each component of MemeCLIP to its performance. The results of our ablation experiments are presented in Table 5. We start with the CLIP ViT-L/14 model and gradually integrate each external module of MemeCLIP. The rise in performance when the projection layers are applied highlights the importance of adapting CLIP’s em-

bedding spaces to our downstream task by disentangling the image and text representations. While the introduction of Feature Adapters initially leads to a temporary dip in F1-Score, it ultimately enables our model to produce more refined image and text representations due to the added learnable parameters. Replacing the linear classifier with a cosine classifier boosts performance by modulating weight updates with a static scaling factor. Finally, Semantic-aware initialization completes MemeCLIP by initializing classifier weights according to the semantic differences in class labels encoded by CLIP’s text encoder, enhancing generalization further.

CLIP	PL	FA	CC	SAI	Acc.	AUC.	F1
✓					72.39±0.61	80.47±1.20	72.33±1.26
✓	✓				74.66±0.03	81.68±0.36	75.04±0.45
✓	✓	✓			75.33±0.17	83.44±0.01	74.77±0.88
✓	✓	✓	✓		75.78±0.32	84.35±0.05	74.92±0.38
✓	✓	✓	✓	✓	76.06±0.23	84.52±0.31	75.09±0.20

Table 5: Ablation experiments performed on MemeCLIP. The results are in the form of *Mean ± Standard Deviation*. PL, FA, CC, and SAI denote Projection Layers, Feature Adapters, Cosine Classifier, and Semantic-Aware Initialization respectively. The last line represents the complete framework. The best results are highlighted in **bold**.

5.4 Comparison with GPT-4

Table 6 compares the performance of MemeCLIP against zero-shot GPT-4 (Achiam et al., 2023) through Microsoft Copilot⁴. We use the prompt “I am trying to build a website with safe images. Can this image be considered safe? Reply with only a number, 0 for yes, and 1 for no.” to manually test GPT-4’s performance on the hate classification task for 100 images from PrideMM’s test set. We qualitatively find that while GPT-4 shows stellar zero-shot performance for hate classification, it tends to make cautious predictions by classifying non-hateful images as hateful or unsafe to the detriment of performance. This behavior may be caused by the stringent safety measures applied to commercial LLMs by LLM vendors (Korbak et al., 2023; Bai et al., 2022).

5.5 Misclassification Analysis

We present two examples of images misclassified by MemeCLIP in Figure 3. The meme presented

Method	Acc.	AUC.	F1
MemeCLIP	73.00	79.06	72.54
GPT-4	68.00	-	69.23

Table 6: Performance comparison between MemeCLIP and the GPT-4 model provided by Microsoft Copilot on the hate classification task.

in Figure 3a is hateful and opposes the values of the LGBTQ+ Pride movement under the guise of benign imagery and text, but is misclassified as non-hateful and neutral. Figure 3b shows a hateful meme mocking an individual, but MemeCLIP classifies it as hate against a community since the text mentions communal words such as “trans” and “women”. Both memes were correctly classified as humorous by the model.

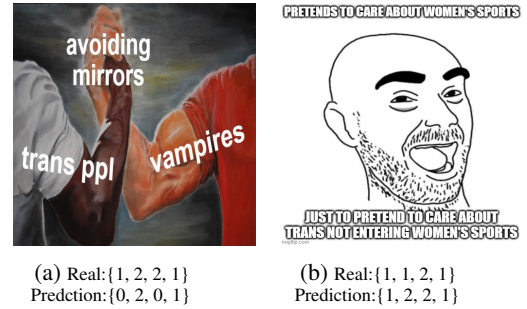


Figure 3: Examples of memes misclassified by MemeCLIP across four tasks. The labels are in the form of {Hate, Target, Stance, Humor}. Label details are outlined in Figure 1.

6 Conclusion

In this work, we release PrideMM, a multimodal and multi-aspect dataset comprising 5,063 memes related to the LGBTQ+ movement, addressing a serious gap in data resources in this domain. This dataset provides memes annotated across three aspects and one sub-class, allowing for greater flexibility in the establishment of ethical guidelines by social media policymakers. We further introduce MemeCLIP, a lightweight yet effective framework to harness CLIP’s knowledge for multimodal meme classification while not relying on extraneous models to create augmented data for training. Our dataset endeavors to foster a deeper understanding of online interaction, community-building, and social change, whereas our framework is a step toward effective content moderation that helps create an inclusive, diverse, and equitable digital environment for all.

⁴<https://copilot.microsoft.com/>

Ethical Considerations

User Privacy. Our dataset only comprises text-embedded images collected from publicly accessible web pages with no inclusion of user data. To the best of our knowledge, there are no copyright concerns associated with them. Since our OCR tool may have potentially scraped user identifiers such as Twitter usernames containing '@', we removed such data from the text during the pre-processing step.

Biases. Due to the size of the dataset and the number of annotation tasks, we acknowledge that some data samples may be misclassified, and not all annotators would agree on the same labels for a given sample. Further, due to the subjective nature of this topic, unintentional biases in data distribution and annotation may exist. We endeavored to minimize such occurrences by using comprehensive annotation guidelines and a three-phase annotation schema.

Potential Risks. PrideMM contains images conveying targeted hate towards specific individuals, communities, ethnic groups, and other entities. While we release our dataset to help robustify content moderation and foster safer spaces online, this data may potentially be used to spread hate and discrimination. We ask researchers to be aware that the inherent biases present in the dataset may negatively influence hate speech detection and moderation methods. Further, we release our multi-aspect dataset in order to give social media policymakers more freedom to censor specific types of harmful memes. However, we acknowledge that this may also lead to over-moderation and negatively affect social media users' freedom of expression.

Annotation. Five annotators were hired to annotate the images for our dataset. The annotators were Indian university students aged 20-25 and were familiar with the LGBTQ+ Pride movement. They were compensated fairly as per the standard local rate. Given the nature of this study, we acknowledged that the annotators may find the images disturbing or distressing. The annotators were given the option to opt out of the annotation process at any given time.

Reproducibility Statement. We provide implementation details and hyperparameter configurations for all the implemented models in Appendix A.1. The PrideMM dataset, source code for MemeCLIP, and MemeCLIP pre-trained weights will be made publicly available through a GitHub repository.

tory.

Environmental Impact. Leveraging hardware such as GPUs to train deep learning models is known to have a significant environmental footprint, primarily due to their high energy consumption resulting in carbon emissions. To mitigate the environmental impact of our research, we adopted a fine-tuning technique for pre-trained models. This method allowed our models to generalize faster on our dataset resulting in fewer computational resources consumed. Once these models are trained using GPUs, they can be loaded on relatively lightweight CPUs for inference purposes, attenuating potential environmental consequences.

Limitations

The proposed dataset PrideMM encompasses memes posted from 2020 to 2024, representing a snapshot of social media interactions only within this specific period, which may fail to capture the dynamics of the LGBTQ+ movement on social media over an extended timeframe. Further, due to the subjective nature of the LGBTQ+ Movement, the size of the dataset, and the number of annotators, the annotation process is inherently prone to biases. While we include images from multiple sources in our dataset, we acknowledge the limited scope of our dataset to these platforms. The intricate nature of memes may not be completely captured by our aspects and more domain-specific aspects may be used to capture the context better. Due to the limited size and availability of labeled datasets in this domain, supervised frameworks such as MemeCLIP may be surpassed by unsupervised methods such as LLMs for content moderation as the capabilities and model size of LLMs progress. Further, the model may exhibit biased predictions due to unintentionally biased data in such datasets.

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⁸<https://github.com/gaopengcuhk/CLIP-Adapter>

Method	Number of Parameters (M)
ViT-L/14	307
CLIP Image-Only	307
CLIP Text-Only	123
CLIP ViT-L/14	428
CLIP-Adapter	428
MOMENTA	358
HateCLIPper	430
ISSUES	452
MemeCLIP	430

Table 7: Number of Parameters for each implemented method.

MOMENTA⁹, HateCLIPper¹⁰, and ISSUES¹¹ to test their respective methods. The total number of parameters for each method is listed in Table 7.

A.2 Data Filtering

We screened the images according to the following criteria:

- **Irrelevant Images:** We curated images relevant to the LGBTQ+ movement and discarded non-relevant images.
- **All Text or no Text Images:** We discarded images that did not contain significant visual content or did not have any embedded text.
- **Non-English Text:** We majorly collected images that had English content, however, some images may contain non-English words. Our OCR tool was set to English ensuring that only English text was extracted from images.
- **Low-Quality Images:** We discard images that were highly distorted, blurred, or degraded. We also removed images with illegible text.

A.3 Annotation Guidelines

In this section, we describe the annotation guidelines used to annotate the dataset. We devise separate guidelines for each of the four tasks.

- **Hate Speech:** This task aimed to identify instances of hate speech in the images. The primary focus was on identifying images that intentionally conveyed hateful sentiments. Annotators needed to distinguish between images expressing strong disagreement without resorting to offensive language and those containing genuine elements of hate speech. This differentiation aimed to guarantee accurate labeling, ensuring that images conveying genuinely hateful sentiment through the visual content, language, or a

combination of both were appropriately identified.

- **Hate Targets:** This task required annotators to identify the targets of hate in hateful images by classifying the images into one of the four classes: *Undirected*, *Individual*, *Organization*, and *Community*. Images were labeled as *Undirected* when they targeted abstract topics, societal themes, or ambiguous targets such as "you", and were not directed toward any specific individuals, entities, or groups. Hateful images targeting specific people including political leaders, celebrities, or activists such as "Joe Biden" and "J.K. Rowling" were annotated as *Individual*. Likewise, the label *Community* was used for instances of images targeting broader social, ethnic, or cultural groups such as "LGBT" or "trans". Lastly, images targeting corporate entities, institutions, or similar organizations such as "Chick-fil-A" and "government" were annotated as *Organization*.
- **Stance:** This task involved annotating the images into either of three distinct categories: *Support*, *Oppose*, and *Neutral*, determined by their context within discussions on the LGBTQ+ movement. The *Support* label was given to images that expressed support towards the goals of the movement, agreed with efforts in fostering equal rights for LGBTQ+ individuals, and promoted awareness for the movement's goals. The *Oppose* label was given to images that conveyed disagreement with the goals of the movement, denied the problems faced by individuals who identified as LGBTQ+, and dismissed the need for equal rights and acceptance. The *Neutral* label was given to images that were contextually relevant to the movement but did not exhibit support or opposition towards the movement.
- **Humor:** In this annotation task, annotators were asked to identify images showcasing humor, sarcasm, or satire related to the LGBTQ+ Pride movement. Annotators were instructed to discern the presence of humor in the images regardless of whether they presented a lighthearted perspective on serious subjects. Note that annotators were asked to annotate images based on whether the creator of the image intended for it to be humorous, and not based on whether the annotator personally found it humorous. This task aimed to capture the nuanced use of text-embedded images for comedic or satirical purposes, thereby enhancing comprehension of the varied expressions within discussions on LGBTQ+.

⁹<https://github.com/LCS2-IIITD/MOMENTA>

¹⁰<https://github.com/gokulkarthik/hateclipper>

¹¹<https://github.com/miccunifi/ISSUES>

Hate		Target			
No Hate	Hate	Undirected	Individual	Community	Organization
brooke (0.919)	woke (0.306)	father (1.111)	rowling (2.122)	transphobes (1.002)	bethesda (2.649)
envy (0.917)	conservative (0.286)	oppression (1.053)	shapiro (1.814)	terfs (0.934)	corporations (2.334)
comfortable (0.859)	children (0.279)	event (1.043)	ben (1.754)	conservatives (0.846)	companies (2.322)
expression (0.842)	warning (0.279)	center (0.932)	biden (1.733)	turning (0.759)	disney (2.265)
subscribers (0.820)	marriage (0.204)	bigot (0.920)	walsh (1.595)	gays (0.758)	russia (2.232)

Table 8: Topic Modeling for Hate and Target classification tasks. We report the top 5 words for every label in each task sorted according to their salience score.

Stance			Humor	
Neutral	Support	Oppose	Humor	No Humor
envy (1.012)	comfortable (1.073)	warning (0.629)	envy (0.480)	risk (0.826)
min (0.964)	subscribers (1.032)	walsh (0.624)	femboy (0.418)	comfortable (0.723)
content (0.938)	brooke (1.000)	matt (0.615)	miss (0.352)	walsh (0.720)
thinks (0.845)	expression (0.946)	replies (0.601)	mematic (0.342)	youth (0.720)
republican (0.804)	supportive (0.907)	oppressed (0.599)	thinks (0.333)	protect (0.640)

Table 9: Topic Modeling for Stance and Humor classification tasks. We report the top 5 words for every label in each task sorted according to their salience score.

A.4 Topic Modeling

We applied the Sparse Additive Generative Models of Text (SAGE) (Eisenstein et al., 2011) topic modeling technique to identify noteworthy words across various class labels within our dataset. We set the hyperparameters max_vocab_size to 1000 and base_rate_smoothing to 1. Tables 8 and 9 present the most notable words for each class as identified by SAGE along with their corresponding salience scores. Among hate targets, words such as "rowling", "shapiro", "conservative", "gays", "bethesda", and "corporations" are assigned high scores by SAGE, helping identify the most targeted entities for each label. Within samples labeled *Support*, words such as "comfortable", "expression", and "supportive" hold relevance as they convey acceptance and support.